Fig1 login page

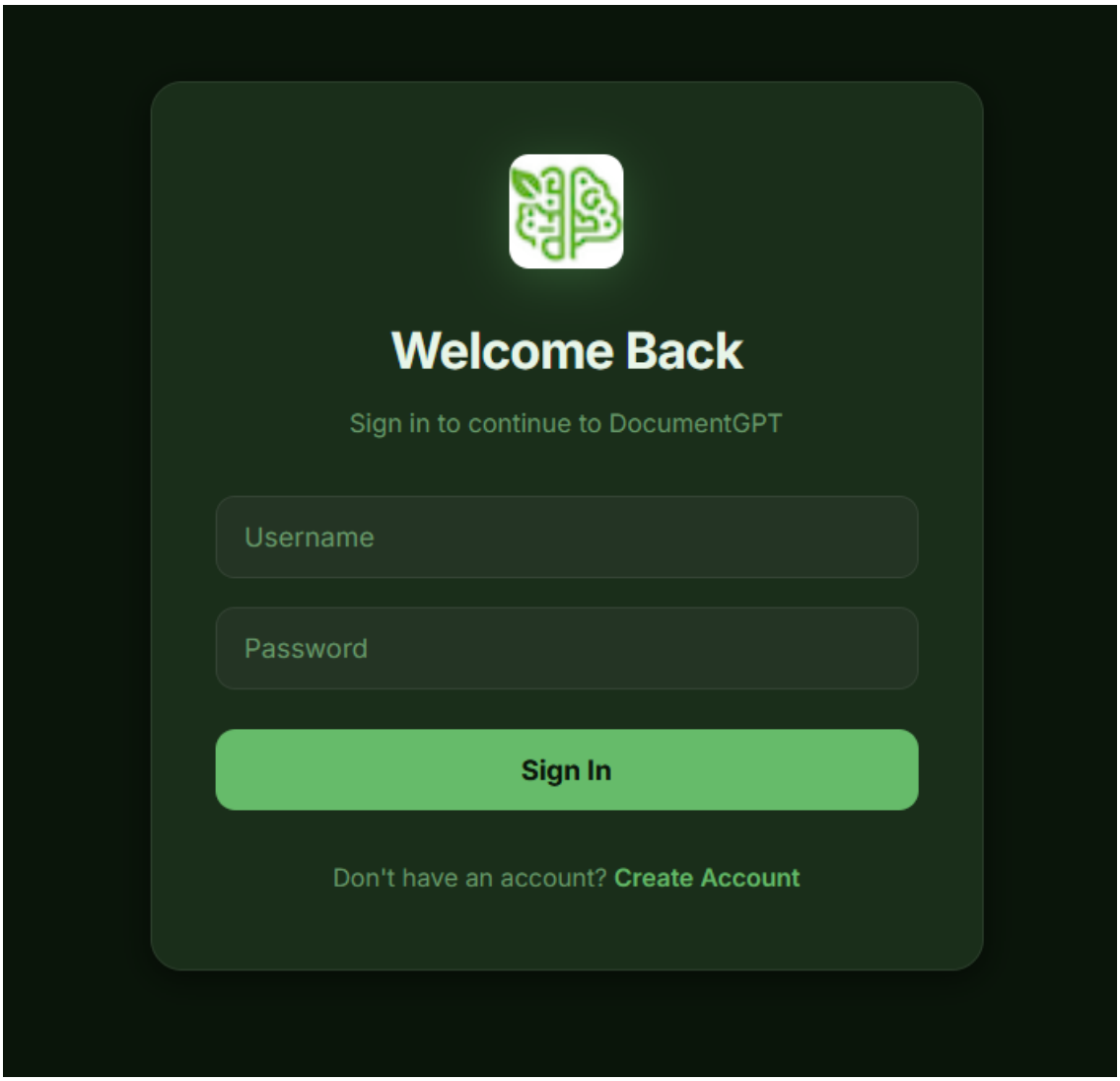


Fig2 registration page

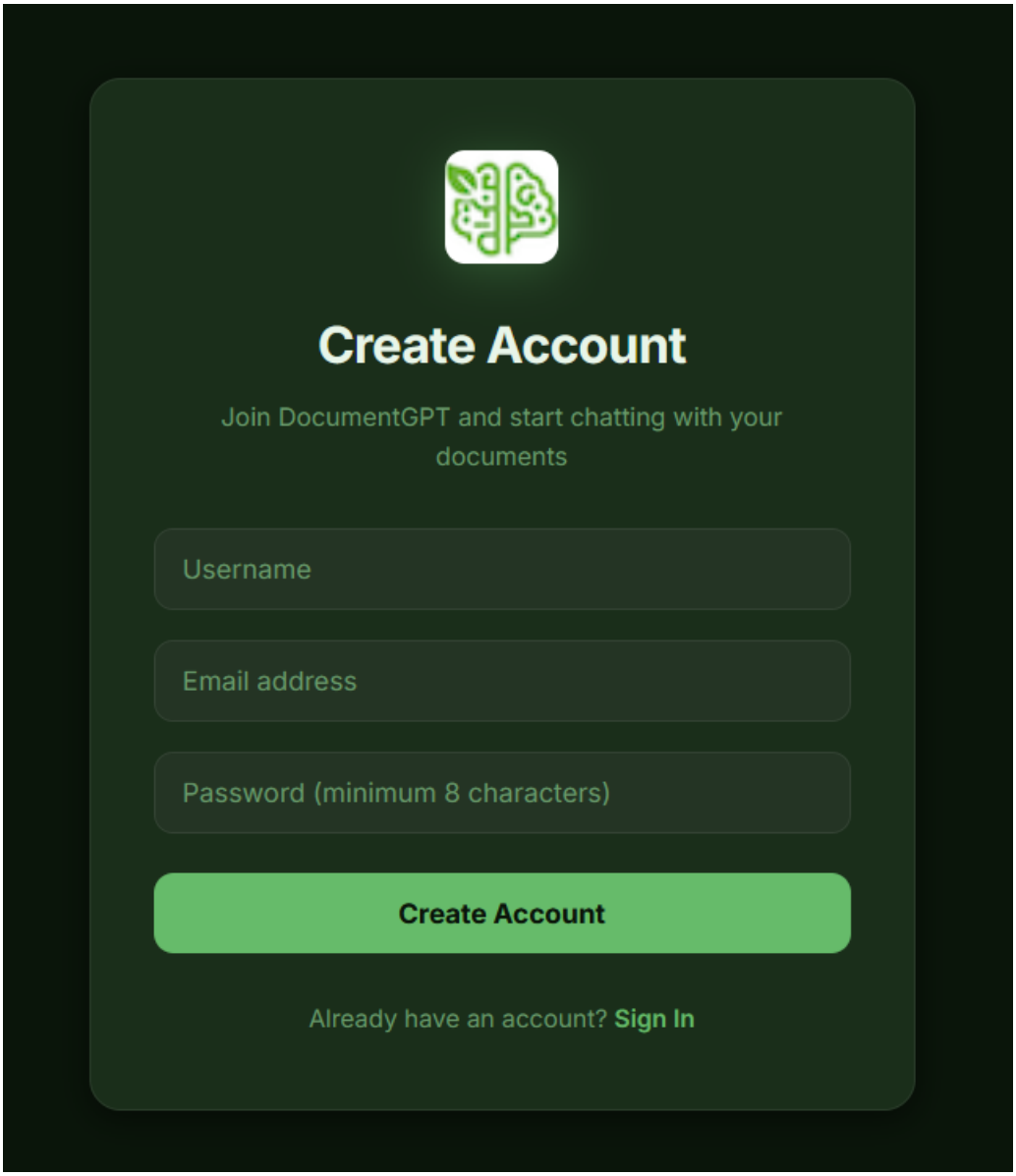


Fig3 document upload

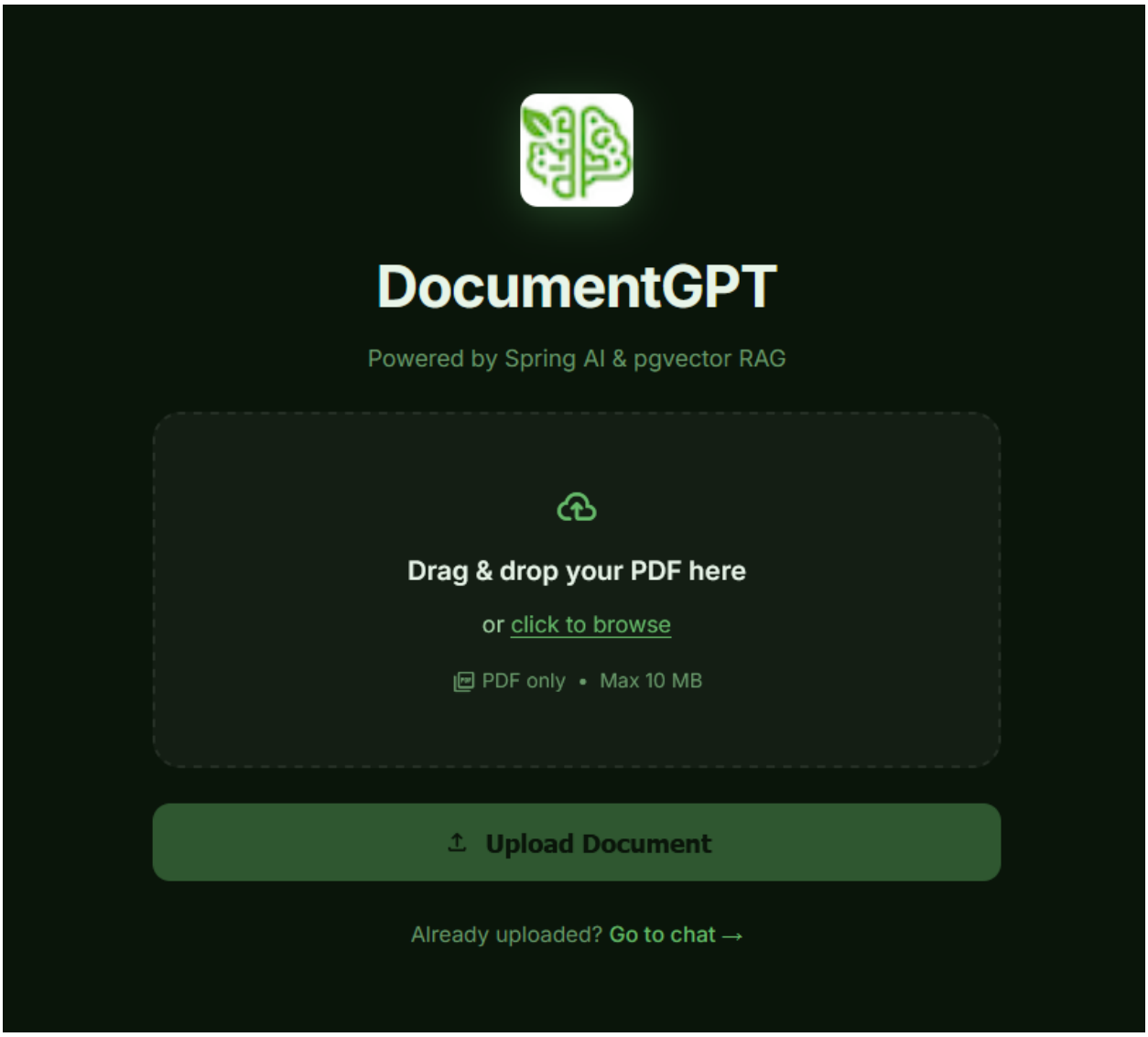


Fig4 Uploaded documents

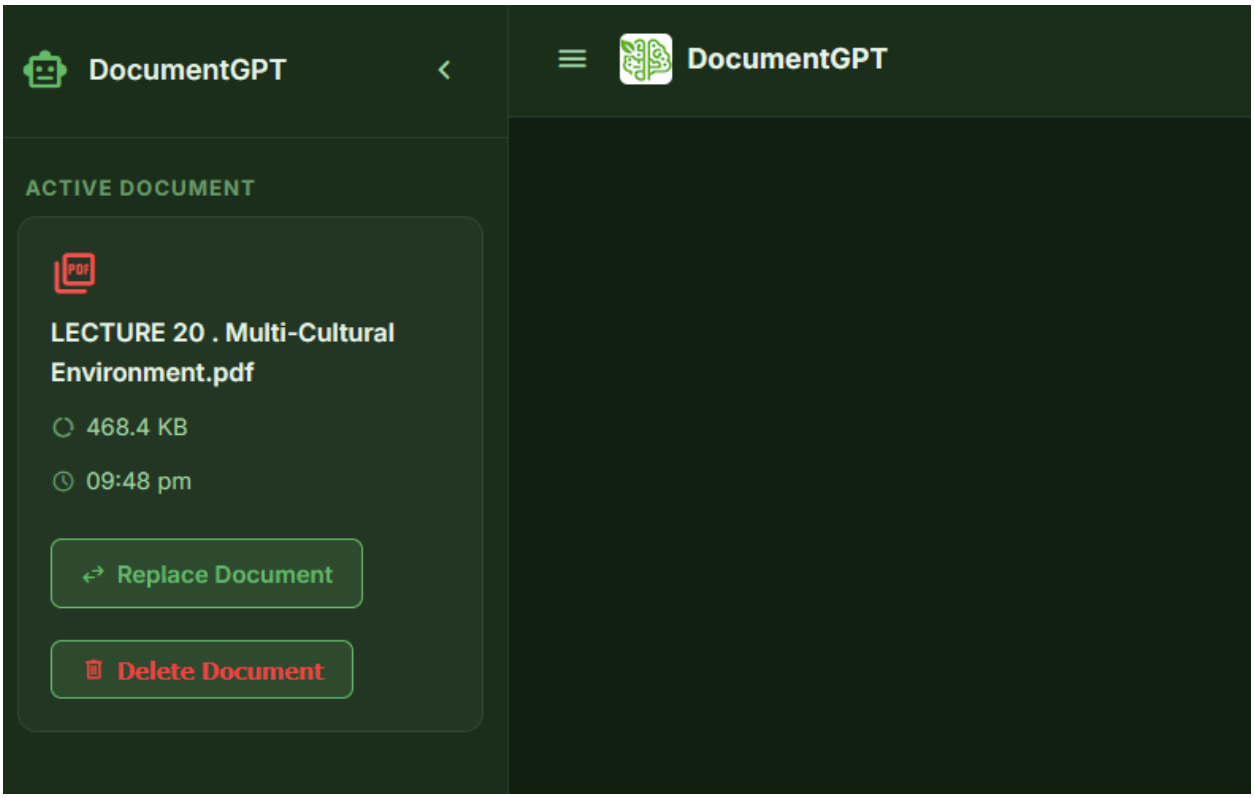


Fig 5 Streaming response

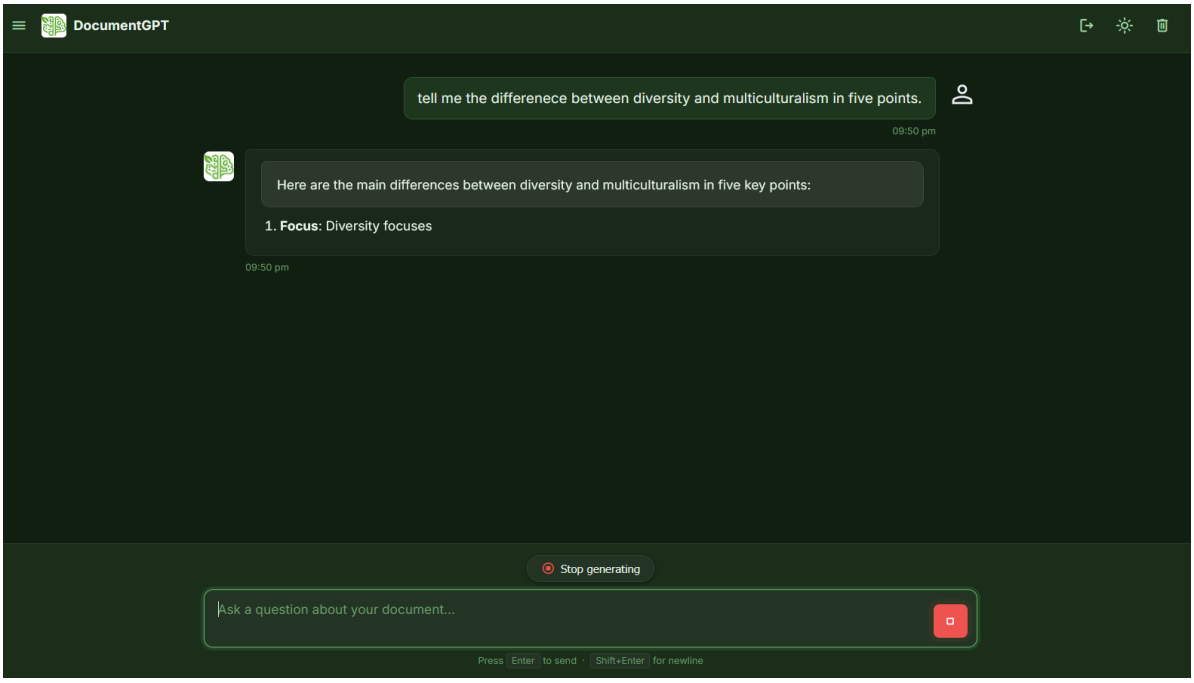


Fig 6 Chat interface with a successful answer

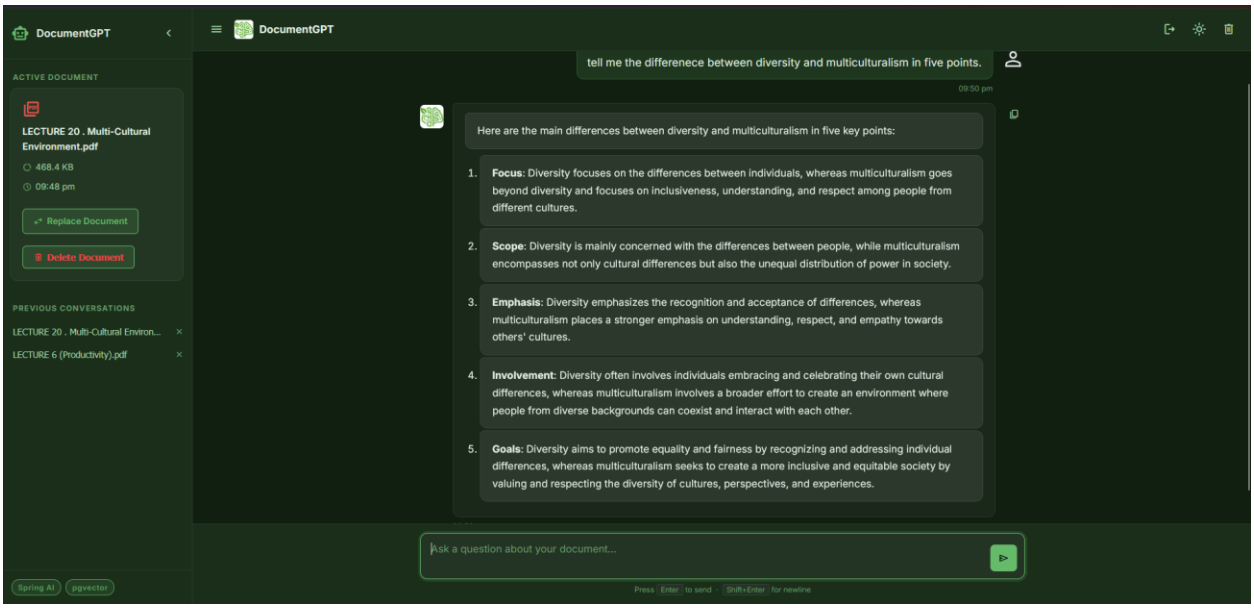


Fig 7 previous saved conversations.

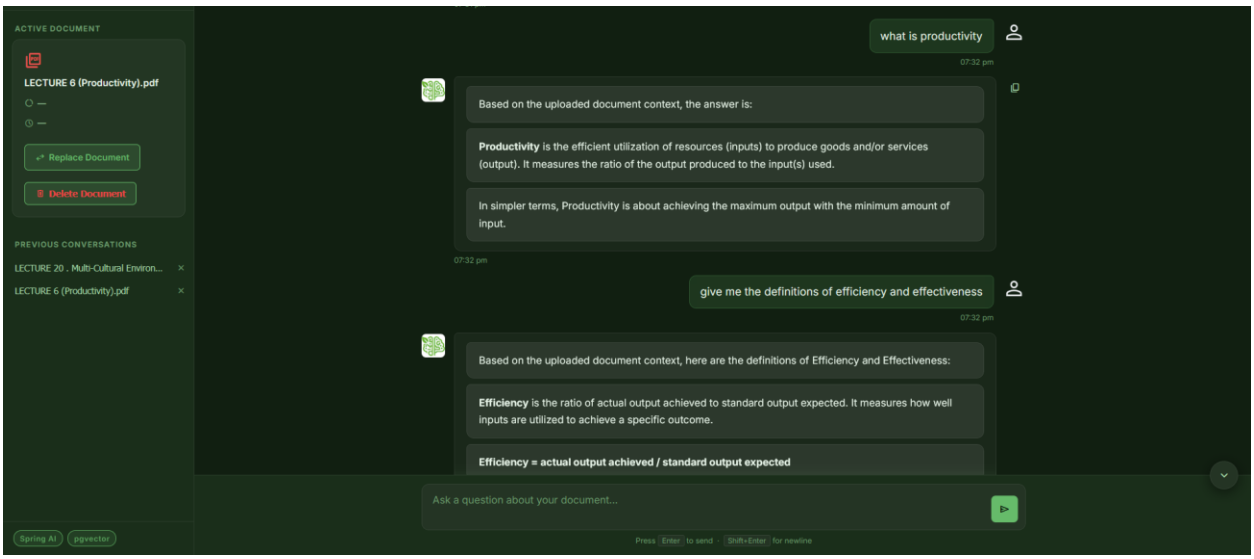


Fig 8 project structure

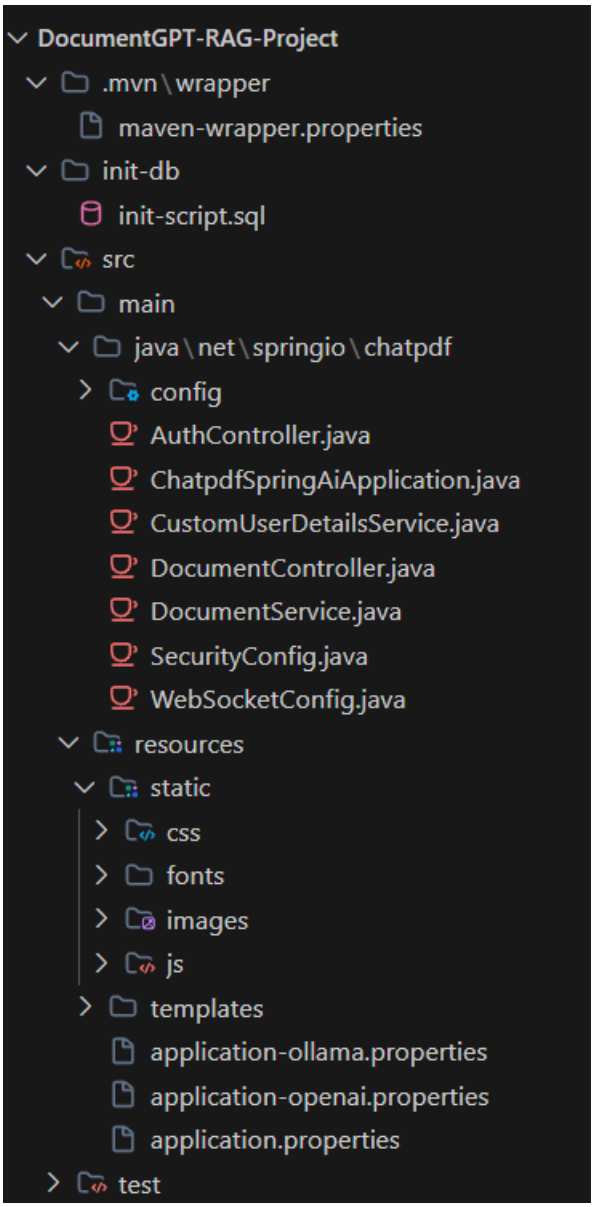
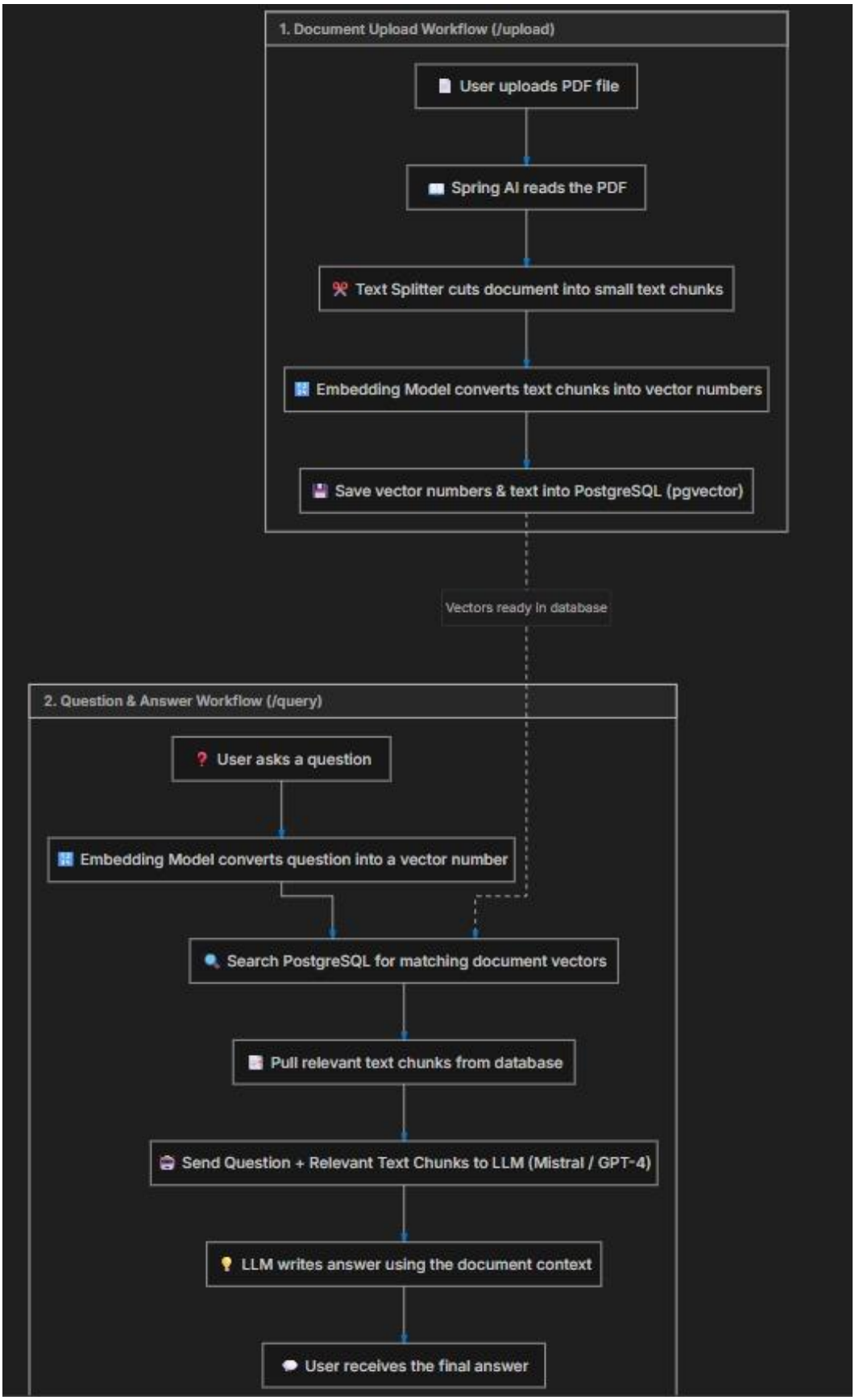


Fig 9 project workflow


1. Document Upload Workflow (/upload)
User uploads PDF file
Spring AI reads the PDF
Text Splitter cuts document into small text chunks
Embedding Model converts text chunks into vector numbers
Save vector numbers & text into PostgreSQL (pgvector)
Vectors ready in database
2. Question & Answer Workflow (/query)
User asks a question
Embedding Model converts question into a vector number
Search PostgreSQL for matching document vectors
Pull relevant text chunks from database
Send Question + Relevant Text Chunks to LLM (Mistral / GPT-4)
LLM writes answer using the document context
User receives the final answer